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Complex Networks

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Introduction

1 | Definitions

In this chapter, we will cover the basic definitions and concepts related to complex networks. We will start with fundamental graph concepts, then explore various network representations, discuss graph isomorphism and permutation, and finally delve into connectedness, distances, subgraphs, planarity, homomorphism, network reduction, and spectral analysis of networks.

1.1 | Basic Graph Concepts

Definition 1.1 (Graph). A graph $G(V, L)$ is a finite nonempty set V of p points (vertexes or nodes) and a set of k couples of distinct points L (links, lines, or edges).

Remark (General Rules).

- *In general, networks have no loops (self-connections).*
- *There are no multigraphs, meaning no multiple links between the same nodes.*
- *We can, however, consider multiple links as weights or as multiple layers (multiplex).*

1.1.1 | Directed and Undirected Graphs

Directed graphs (DiGraphs) are graphs with directed edges, known as arcs, indicating a “from-to” relationship. They are used to describe hierarchical relationships, fluxes, or the passing of messages and goods. Undirected graphs contain only symmetric links and are used to describe symmetrical relations such as friendship, chemical bonds, or statistical correlation. Note that directed graphs can be symmetrized; while this loses some information, it allows the use of simpler properties and algorithms.

1.1.2 | Walk, Trail, Path, and Cycle

Let us consider a graph with nodes i and j . We can define different types of connections between these nodes:

- **Walk:** A node and edge sequence between two nodes.
- **Trail:** A walk with all distinct links.
- **Path:** A walk with all distinct nodes and links.
- **Cycle:** A closed path with $N_c \geq 3$ nodes.

This classification is important for understanding the structure of networks and for defining concepts such as connectedness and distances. Different algorithms are used to find walks, trails, paths, and cycles in a graph, which can be crucial for applications like network traversal, routing, and community detection.

1.2 | Network Representations

It is important to note that a graph can be represented in various ways, each with its own advantages and disadvantages. The choice of representation often depends on the specific application and the properties of the graph being studied.

1.2.1 | Adjacency Matrix

The most common representation of a graph is the adjacency matrix, which is a square matrix that encodes the presence or absence of links between nodes. For unweighted graphs, the entries are binary (0 or 1), while for weighted graphs, the entries can take on a range of values corresponding to the weights of the links.

Definition 1.2 (Adjacency Matrix). The adjacency matrix A is an $N \times N$ matrix representing N nodes. For unweighted networks, $A_{ij} = 1$ indicates a topological link from node i to node j , with $A_{ii} = 0$ to indicate no loops. For weighted networks, $A_{ij} = w_{ij}$ where w_{ij} is the weight of the link.

Directed Graph

Consider the following directed graph with 6 nodes, we can represent it with the following adjacency matrix:

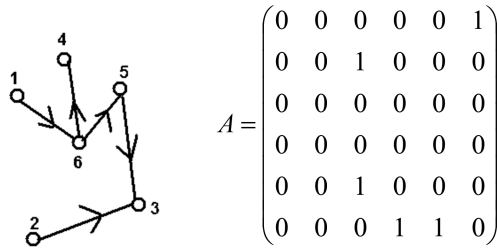


Figure 1.1: A directed graph and its corresponding non-symmetric adjacency matrix.

Undirected Graph

Consider the following undirected graph with 6 nodes, we can represent it with the following adjacency matrix:

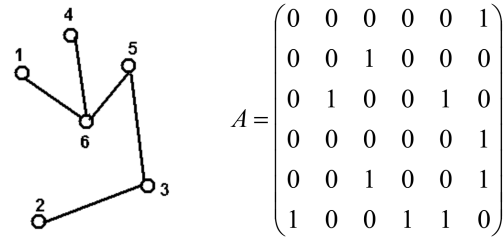


Figure 1.2: An undirected graph and its corresponding symmetric adjacency matrix.

Remark (Logical Operators on Adjacency Matrices). We can apply element-wise logical operators on adjacency matrices:

- Network symmetrization uses a logical OR: $A' = \max(A, A^T) = A \cup A^T$.
- Identifying reciprocal links uses a logical AND: $A'' = \min(A, A^T) = A \cap A^T$.
- Identifying strictly directed links uses logical XOR: $XOR(A, A^T)$.

1.2.2 | Weighted Network

In a weighted network, the adjacency matrix contains values that represent the strength of the connections between nodes:

$$A_{ij} = w_{ij} \text{ link from node } i \text{ to node } j \text{ with weight } w_{ij}.$$

This can be interpreted in various ways depending on the context, such as the number of interactions, the intensity of relationships, or the capacity of links.

It can be seen as a representation of a **multigraph** with N multi-edges: $w_{ij} = N$. If all weights are positive, we have useful theorems for matrices with positive values (Perron-Frobenius theorem) and we can connect to "physical" situations (transition probabilities of diffusion processes). Negative weights can be meaningful too: in **signed networks** the sign of the weight can represent different types of relationships or interactions, such as:

- **Friend-enemy**: in social networks, positive weights can represent friendships while negative weights can represent enmity.
- **Attraction-repulsion**: in physical systems, positive weights can represent attractive forces while negative weights can represent repulsive forces.
- **Ferromagnetic-antiferromagnetic**: in spin systems, positive weights can represent ferromagnetic interactions while negative weights can represent antiferromagnetic interactions.

One can also operate a weighted matrix symmetrization:

- Average symmetrization: $A' = \frac{1}{2}(A + A^T)$.
- Random matrix symmetrization: $A' = \frac{1}{\sqrt{2}}(A + A^T)$.

1.2.3 | Incidence Matrix

We can also represent a graph using an incidence matrix, which is an alternative to the adjacency matrix. The incidence matrix is particularly useful for certain types of analyses, such as flow problems and network dynamics.

Definition 1.3 (Incidence Matrix). The incidence matrix I of a graph $G(V, L)$ is an $L \times V$ matrix (number of links by number of vertexes). It contains -1 for outgoing connections and $+1$ for ingoing connections:

$$\text{if } A_{ij} \neq 0, \quad I_{Ni} = \mp 1, \quad I_{Nj} = \pm 1.$$

It can be interpreted as a "generalized derivative" on the graph, giving the oriented difference between values on adjacent nodes.

Thus the incidence matrix can be used to analyze the flow of information or resources through the network, and it is particularly useful for studying directed graphs and their properties. The incidence matrix can also be used to derive the adjacency matrix through the relation $A = I^T I - D$, where D is a diagonal matrix that accounts for the degree of each node.

Generalized Derivative

The incidence matrix can be interpreted as a generalized derivative on the graph, giving the oriented difference between values on adjacent nodes. The incidence matrix allows us to compute the net flow into or out of each node, which is essential for understanding the dynamics of processes.

Given a vector of values $\mathbf{f}_i$ for each node, the incidence matrix I applied on the vector gives:

$$\sum_j I_{ij} \cdot \mathbf{f}_j = \Delta_i,$$

$$\Delta_i = \mathbf{f}_i - \mathbf{f}_j \forall \text{adjacent nodes},$$

where Δ_i represents the net flow into or out of node i based on the values of its neighbors. This interpretation is crucial for analyzing diffusion processes, random walks, and other dynamic phenomena on networks.

Geometrically, a network acts as a spatial structure: while a regular lattice discretizes Euclidean space, a complex graph represents a more intricate manifold. However, one must be careful, as the properties of this discrete manifold do not necessarily hold in the continuous limit.

1.2.4 | Sparse Representations

For sparse graphs where the number of links is much smaller than V^2 , other representations are preferred:

- **Edge Matrix:** An $L \times 2$ matrix that stores single links instead of the full matrix.
- **Adjacency List:** Each line describes a node and contains the indices of its arrival nodes. This is particularly useful for diffusion processes, such as simulating random walks on a graph.

1.3 | Graph Isomorphism and Permutation

Definition 1.4 (Graph Isomorphism). An isomorphism is a 1-1 correspondence between vertexes that preserves the link structure. It corresponds to a node index permutation.

Remark (Re-indexing). *Node permutation represents node reordering. Re-indexing can be based on a priori information (like intrinsic metrics) or network-based centrality measures, and is very important for network visualization.*

Example (Protein and DNA Contact Maps). A network representation of peptide bonds serves as a discretization of a 3D distance matrix. Spatial closeness acts as an “intrinsic” metric for the graph.

1.4 | Connectedness

1.4.1 | Undirected Graphs

An undirected graph is connected if it is not divided into two or more non-communicating parts. Algebraically, this means there is no permutation that reduces the adjacency matrix into diagonal blocks.

1.4.2 | Directed Graphs

In a directed graph, not every node is necessarily reachable from any other. Connectedness is classified into three types:

- **Strongly connected:** Every node is connected to another by a path, implying the existence of cycles.
- **Unilaterally connected:** For every two nodes, there is at least one path.
- **Weakly connected:** Every node is connected to another by at least a semipath.

Remark (Block Matrix Reduction). *For directed graphs, the adjacency matrix can sometimes be reduced to blocks where nodes in block C are reachable from block A , but not vice versa. In such cases, block C acts like a “sink” or “absorbing state” in transition processes.*

1.5 | Distances and Shortest Paths

1.5.1 | Node Distance

The distance between two nodes is defined as the shortest path between them. In weighted networks, nodes that are “closer” have a greater weight, and distance can be calculated using harmonic composition (similar to capacitors in series). The larger the weight, the smaller the distance.

1.5.2 | Shortest Path Algorithms

The distance between two nodes equals the minimum number of links (geodesic) between them. The diameter of a graph is the maximum path length, while the average path length is the average over all paths. These are typically calculated using Dijkstra’s algorithm (one-to-all) or the Floyd-Warshall algorithm (all-to-all).

1.6 | Subgraphs and Special Network Topologies

1.6.1 | Subgraphs and Cliques

Definition 1.5 (Induced Subgraph e K-Clique). • **Subgraph:** A subset of a graph's nodes and links.

- **Induced Subgraph:** A subset of the nodes of G with ALL the links of G that exist between those specific nodes.
- **K-Clique:** A fully connected subgraph with degree K .

1.6.2 | Complete Graphs

A completely connected graph K_N has all possible links, meaning every node has a degree of $K = N - 1$. The graph complement has the same nodes but the “opposite” links.

1.6.3 | Bipartite Graphs

Definition 1.6 (Bipartite Graph). A graph that connects two different types of nodes between each other, but NOT between themselves. It is represented by an $M \times N$ rectangular adjacency matrix. In terms of matrix algebra, a bipartite graph is an imprimitive matrix of order 2.

Remark (Tensor Contraction). *Given a rectangular adjacency matrix B representing a bipartite graph, we can obtain two square networks connecting nodes of the same type via contraction: $B \cdot B^T$ and $B^T \cdot B$. Nodes of the same type are connected if there is at least one path between them through a node of the other type.*

Example (The Human Disease Network). An example of a bipartite graph is the “disease” (e.g., OMIM database), a gene-pathology network. By contracting the matrix, we can obtain a gene-gene network based on shared diseases, or a pathology-pathology network based on shared genes.

1.6.4 | Trees, Stars, and DAGs

Remark (DAG Adjacency Matrix).

1.7 | Planarity and Homomorphism

1.7.1 | Planarity and Genus

1.7.2 | Graph Homomorphism

1.8 | Network Reduction and Filtering

1.8.1 | Minimum Spanning Tree (MST)

Definition 1.7 (Min/Max Spanning Tree).

1.8.2 | Link Selection Rules

1.9 | Spectral Analysis of Networks

1.9.1 | The Eigenvalue Spectrum

1.9.2 | Powers of the Adjacency Matrix

1.9.3 | Spectra of Specific Graphs

1.9.4 | Isomorphism and Co-spectrality

Remark (Co-spectrality).

1.9.5 | Graph Distances

2 | Random Graph Ensembles

In this chapter, we will explore the concept of random graph ensembles, which are fundamental in understanding the properties and behaviors of complex networks. We will start with the simplest model, the Erdős-Rényi random graph, and then move on to more complex models such as the Stochastic Block Model (SBM). We will also discuss the thermodynamic limit, degree distribution, phase transitions, and other properties of these random graph ensembles.

2.1 | Erdős-Rényi Random Graphs

We are going to start with the simplest model of random graphs, which is the Erdős-Rényi (E-R) model. This model serves as a fundamental null model for understanding the properties of complex networks.

Definition 2.1 (Erdős-Rényi Random Graphs). Networks whose interactions (links) between nodes are chosen randomly, without a priori correlations or rules. In statistical mechanics terms, it is comparable to a “perfect gas” with molecules (nodes) experiencing random “sparse” impacts (links).

It can be seen as a **null model** for complex networks, as it represents a system with no structure or correlations.

If we expect to find a certain property in a real network, we can compare it to the expected value of that property in an E-R random graph. If the observed value significantly deviates from the expected value, we can infer that there is some underlying structure or correlation in the real network that is not captured by the E-R model. It can also be used to confront different systems, such as comparing the structure of a social network to that of a biological network, to identify unique features and commonalities.

If we make the statistical analogy with a gas, the E-R model represents a perfect gas, where the nodes are like molecules that interact randomly without any specific rules or correlations. The edges, which are generated randomly, represent some average quantities of the system, not the distance between nodes or the strength of interactions. The E-R model is a useful starting point for understanding the properties of complex networks, but it is important to note that real-world networks often exhibit more complex structures and correlations that are not captured by this simple model.

2.2 | Statistical Ensembles of Networks

If we imagine to take a picture of our ideal gas system, we would see a random distribution of molecules (nodes) and their interactions (links, diluted interactions, since we are speaking of ideal gas). The E-R model captures this randomness and serves as a baseline for understanding the properties of complex networks. In the next instant, the picture would change, and we would see a different random distribution of molecules and interactions, but the overall properties of the system would remain the same, as they are determined by the average quantities rather than the specific configuration of nodes and links. We will see that these analogies with ensembles will be very useful to understand the properties of complex networks, and also the E-R model will maintain some properties constant even if we change the specific configuration of nodes and links, which is a key feature of random graph ensembles.

We are going to use **ensembles of networks** with fixed parameters: same macroscopic state (e.g. link density, degree pdf) different microstates (single-node properties), and we will see that the properties of the ensemble are determined by the average quantities rather than the specific configuration of nodes and links. The definition of the ensemble will suggest a construction rule to generate the microstates of the ensemble and to compute the probability of generating a specific microstate.

Remark. *The number of ALL symmetric networks (microstates) with N nodes (whatever the links) is:*

$$2^{\frac{N(N-1)}{2}}, \quad \sum_{i=1}^N \binom{N}{i} = 2^N.$$

This means that the number of possible configurations of links between N nodes is exponential in the number of nodes, which highlights the complexity of the space of possible networks and the importance of using ensembles to understand their properties.

The most common ensembles are the microcanonical ensemble, where we fix the number of links, and the canonical ensemble, where we fix the probability of having a link between two nodes. Let us introduce these two ensembles in more detail.

2.2.1 | Microcanonical Ensemble $G_{N,M}$

Definition 2.2 (Microcanonical Ensemble). An ensemble of networks with N nodes, representing a fixed system size, and M links, representing the system energy. The links are randomly selected from the possible $N(N-1)/2$ for a symmetric network or $N(N-1)$ for a directed graph.

The idea behind a microcanonical ensemble is that we are looking at all possible configurations of a network with a fixed number of nodes and links. Each configuration (or microstate) is equally likely, and the properties of the ensemble are determined by the average quantities rather than the specific configuration of nodes and links.

Looking at the properties of the microcanonical ensemble, we can see that it is a very simple model, but it can still capture some important properties:

- The N nodes represent the system size, which is fixed.
- The M links represent the system energy, which is also fixed.

- The links are randomly selected from the possible $N(N - 1)/2$ for a symmetric network or $N(N - 1)$ for a directed graph, which means that there is no specific structure or correlation in the network.

So we have

$$M_{max} = \frac{N(N - 1)}{2}, \quad \#G_{N,M} = \binom{M_{max}}{M},$$

such that for a fixed number of nodes N and links M , the number of possible configurations (or microstates) of the network is given by the binomial coefficient $\binom{M_{max}}{M}$, which counts the number of ways to choose M links from the total possible M_{max} links. This highlights the combinatorial nature of the microcanonical ensemble and the importance of using statistical methods to understand its properties.

Remark (Construction Rule). *The construction rule of one instance of the ensemble is:*

1. Take N nodes.
2. Repeat M times: choose 2 nodes randomly and connect them, checking for loops or existing links.

The probability of generating “bad” links, such as loops and multilinks, is about $1/N$. The effect of this selection becomes negligible for large N .

The probability of generating a loop is $1/N^2$, and the probability of generating a multilink is M/N^2 . In order to connect two times the same nodes, the probability is $1/N^2$. For large N , i.e. in the thermodynamic limit, these probabilities become negligible, and the microcanonical ensemble can be effectively generated by randomly selecting M links from the possible $N(N - 1)/2$ links without worrying about loops or multilinks.

2.2.2 | Canonical Ensemble $G_{N,P}$

Often it is more convenient to **fix the probability** of having a link between two nodes, rather than the number of links. This leads us to the canonical ensemble.

Definition 2.3 (Canonical Ensemble). An ensemble of networks with N nodes (fixed system size) and a fixed probability P to have a link between them, representing a fixed average energy or temperature.

This means that in the canonical ensemble, given the nodes and the probability P , we can generate a random graph by connecting each pair of nodes with a link with probability P . The properties of the ensemble are determined by the average quantities:

$$P = \langle E \rangle \propto T, \quad \langle l \rangle = P \frac{N(N - 1)}{2}.$$

Binomial Link Distribution

Given N and P I do not have a fixed number of networks belonging to the canonical ensemble Better: ALL graphs with N nodes belong to $G_{N,P}$ with a different probability depending on P The probability of having a graph with m links and N nodes is governed by the binomial distribution:

$$P(m) = \binom{N(N - 1)/2}{m} P^m (1 - P)^{N(N - 1)/2 - m}$$

The average grows as the ...

Remark (Construction Rule). *The probability of having a graph with m links and N nodes is governed by the binomial distribution. To construct an undirected graph:*

- Take $N(N-1)/2$ values A_{ij} uniformly distributed between 0 and 1.
- If $A_{ij} > 1 - P$ then $A_{ij} = 1$, else $A_{ij} = 0$.
- Apply symmetrization using logic “OR”: $A \vee A^T$.

If the network is sufficiently large, the probability of having a graph with m links and N nodes is governed by the binomial distribution, which can be approximated by a Poisson distribution in the limit of large N and small P with $P \cdot N$ kept constant. The average number of links in the canonical ensemble is given by $\langle l \rangle = P \frac{N(N-1)}{2}$, which grows as the square of the number of nodes for fixed P . This highlights the fact that in the canonical ensemble, the number of links can vary widely depending on the specific configuration of nodes and links, but the average properties of the ensemble are determined by the probability P and the number of nodes N .

2.2.3 | Equivalence of Ensembles

For $N \gg 1$, the canonical link distribution tends to a Dirac’s delta function. All instances of the canonical ensemble tend to have a number of links equal to the average link number, thus recovering the microcanonical property.

[...]

2.2.4 | Degree Probability Density Function (pdf)

The probability of having a node with k links is the probability of randomly choosing k links among $N-1$:

$$p(k) = \binom{N-1}{k} P^k (1-P)^{N-1-k},$$

Balls and Holes

If we consider the problem of distributing k links (balls-couples) among $N-1$ possible connections (holes), we can see that the degree distribution follows a **binomial distribution**, which can be approximated by a Poisson distribution in the limit of large N and small P with $P \cdot N$ kept constant. This analogy helps us understand the degree distribution in random graphs and how it arises from the random process of connecting nodes with links.

The degree distribution in the canonical ensemble is given by the binomial distribution, which can be approximated by a Poisson distribution in the limit of large N and small P with $P \cdot N$ kept constant

$$p(k) = \binom{N-1}{k} P^k (1-P)^{N-1-k}$$

The average node degree $\langle k \rangle$ is given by $P(N-1)$, which grows linearly with the number of nodes for fixed P :

$$\langle k \rangle = \frac{\langle \# \text{links} \rangle}{N} = \frac{2P \frac{N(N-1)}{2}}{N} = P(N-1) \approx PN,$$

while the average link density is

$$p \frac{N(N-1)}{2}.$$

This highlights the fact that in the canonical ensemble, the degree distribution can vary widely depending on the specific configuration of nodes and links, but the average properties of the ensemble are determined by the probability P and the number of nodes N .

Remark (Bernoulli to Poisson to Delta). *In the thermodynamic limit ($N \rightarrow \infty$, $P \rightarrow 0$) with $P \cdot N = \lambda$ kept constant, the distribution becomes a Poisson distribution:*

$$p(k) \approx \frac{\lambda^k}{k!} e^{-\lambda}$$

In this limit, a random Bernoulli graph tends to a regular graph with equal connectivity for all nodes, even if node proximity is not regular as in a lattice. The Erdős-Rényi model tends to an infinite-dimensional regular lattice.

$$P \cdot N = \lambda = \text{constant}(P \rightarrow 0, N \rightarrow \infty),$$

$$p(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad \langle k \rangle = \lambda, \quad \sigma = \sqrt{\lambda},$$

$$\lim_{\lambda \rightarrow \infty} p(k) = N(\lambda, \sqrt{\lambda}) \rightarrow \delta(x - \lambda).$$

[...]

2.3 | Phase Transitions and the Giant Component

Considering the probability of having a property in an ensemble with N nodes

$$P = P(N),$$

depending on the trend we get behaviors that lead to have certain properties valid for almost all graphs or almost none. This is the idea of a phase transition, where we have a critical point that separates two different phases of the system. In the context of random graphs, one of the most important properties is the emergence of a giant component, which is a connected component that contains a finite fraction of the nodes in the graph. The emergence of a giant component is a phase transition that occurs at a critical value of the average connectivity λ .

2.3.1 | Average Connectivity

In E-R networks, the *threshold for transitions depends on the network size N and the connection probability p* . We use the average connectivity λ as the control parameter:

$$\langle k \rangle \approx pN = \lambda$$

2.3.2 | The Giant Component

Definition 2.4 (Giant Component Emergence). • For $\lambda < 1$, there is no giant component.

- For $\lambda \geq 1$, there is a giant component that, for increasing λ , tends to include all the nodes.

Remark (Transcendental Equation). *If we define u as the probability of a node not belonging to the Giant Component, the average field equation yields the transcendental equation:*

$$u = e^{\lambda(u-1)}$$

The order parameter S , which is the fraction of nodes belonging to the giant component, is $S = 1 - u$.

... GC order parameter...

One can find similarities with the **percolation model**: If we have a lattice of nodes and we randomly occupy the links with probability p , we can see that there is a critical value of p above which a giant component emerges, which is similar to the phase transition in random graphs. The percolation model can be seen as a special case of the random graph model, where the nodes are arranged in a lattice and the links are occupied with a certain probability.

Adding one link at a time creates larger and larger clusters, which is known as the “**avalanche phenomenon**”. This is similar to the process of percolation, where adding one link can lead to the emergence of a giant component that contains a finite fraction of the nodes in the graph: after a critical point, the network becomes traversable, connected. The phase transition in random graphs can be understood as a percolation process, where the control parameter is the average connectivity λ .

2.4 | Network Properties in Random Graphs

We can expect that in a random graph, the properties of the network will be determined by the average quantities rather than the specific configuration of nodes and links. This means that we can use statistical methods to understand the properties of random graphs and how they arise from the random process of connecting nodes with links.

2.4.1 | Diameter and Average Path Length

Assuming a regular graph ($N \gg 1$) and a sparse network (regular tree, no cycles, so that from a node there is a unique path to any other node and a very small probability of coming back to the starting node) with

$$\#l \sim O(N) = o(N^2),$$

we can define the **diameter** of the graph as the longest shortest path between any two nodes.

Starting from a node, one can reach $\langle k \rangle$ nodes in one step, $\langle k \rangle^2$ nodes in two steps, and so on. After d steps, one can reach $\langle k \rangle^d$ nodes. Therefore, the diameter of the graph can be estimated by solving the equation:

$$\langle k \rangle^d \sim N.$$

In a random graph, the diameter grows logarithmically with the number of nodes, which means that even for large networks, the distance between nodes remains relatively small. This is a key feature of random graphs and is often referred to as the “small-world” property.

Let us give some insight into the solution of that equation. We can take the logarithm of both sides to get:

$$\dots$$

A similar result holds for the **average path length**

$$\langle l(N) \rangle \propto \ln N.$$

As the network size grows, the distance between nodes grows slowly, which for a D-lattice, translates in the diameter growing as $N^{1/D}$, which is much faster than logarithmic growth.

Random networks are **traversable**, meaning that there is a path between any two nodes, and the distance between nodes grows logarithmically with the number of nodes. This is a key feature, a trivial property of random graphs and is often referred to as the “**small-world**” property, which has important implications for the dynamics and functionality of complex networks.

2.4.2 | Degree Correlations

Let us study the relationship between the degrees of neighboring nodes in a random graph. In an E-R random graph, there is no correlation between the degrees of neighboring nodes, which means that the degree of a node does not depend on the degree of its neighbors. This is because the links are generated randomly, and there is no specific structure or correlation in the network. If the nodes are correlated, we would expect to see a relationship between the degrees of neighboring nodes, such as a positive correlation (where high-degree nodes tend to be connected to other high-degree nodes) or a negative correlation (where high-degree nodes tend to be connected to

low-degree nodes). However, in an E-R random graph, there is no such correlation, and the degree of a node is independent of the degree of its neighbors.

There is an **absence of correlations** between node features. Specifically, there is ALMOST none for finite-size graphs and none in the thermodynamic limit ($N \rightarrow \infty$ and $\#l = O(N)$). For undirected graphs:

$$p(k_1, k_2) \approx p(k_1) \cdot p(k_2).$$

In-Degree and Out-Degree Correlations

Similarly, in an ER directed network, there is no correlation between in-degree and out-degree: $p(k_{IN}, k_{OUT}) = p(k_{IN}) \cdot p(k_{OUT})$. This means that the in-degree of a node does not depend on its out-degree, and vice versa. This lack of correlation is a key feature of random graphs and distinguishes them from more structured networks, where there may be correlations between the degrees of neighboring nodes.

2.5 | Stochastic Block Model (SBM)

Another important model of random graphs is the Stochastic Block Model (SBM), which is a more complex null model that can modelize a nontrivial community structure of the network. It can be obtained by combining diagonal E-R blocks, possibly of different sizes and link densities, with non-diagonal blocks of random links. The diagonal blocks represent the communities, where nodes within the same community are more likely to be connected, while the non-diagonal blocks represent the connections between different communities, which can be random or have a different link density.

Definition 2.5 (Stochastic Block Model). A more complex null model that can modelize a non-trivial community structure of the network. It can be obtained by combining diagonal E-R blocks, possibly of different sizes and link densities, with non-diagonal blocks of random links.

The main algorithm backbone is the same to generate symmetric or directed E-R networks of specific size and link density. The non-diagonal blocks connecting communities of different size (e.g. with N and M nodes) will be rectangular ($N \times M$).

What makes the SBM more complex than the E-R model is that it can capture the community structure of real-world networks, where nodes are organized into groups or communities that have a higher density of connections within the group than between groups. This makes the SBM a more realistic model for understanding the properties of complex networks, as many real-world networks exhibit community structure.

We can play with the structure of the blocks to generate different types of networks, such as assortative networks (where nodes within the same community are more likely to be connected) or disassortative networks (where nodes within the same community are less likely to be connected). We can model networks with a core-periphery structure, where there is a dense core of nodes that are highly connected to each other and a periphery of nodes that are less connected. We can also model networks with a hierarchical structure, where there are multiple levels of communities nested within each other. The SBM is a powerful tool for understanding the properties of complex networks and can be used to generate synthetic networks that mimic the structure of real-world networks.

3 | Spectral Analysis of Random Graphs

We are going to analyze the spectral properties of random graphs, which are important for understanding the dynamics and functionality of complex networks. We will start with the spectral density of the adjacency matrix, which is a key tool for understanding the structure of random graphs. We will then explore the eigenvalue distribution and its implications for the stability and dynamics of random graphs.

The **observables** of interest in the study of random graphs include

- Eigenvalue distribution of the adjacency matrix, which can provide insights into the structure and dynamics of the network.
- Maximum eigenvalue, which can be related to the stability of the network and the presence of a giant component.
- Nearest neighbor spacing distribution (NNSD), which can provide insights into the correlations between eigenvalues and the presence of community structure in the network.

These spectral properties can be analyzed using tools from random matrix theory, which provides a framework for understanding the statistical properties of eigenvalues in large random matrices. By studying the spectral properties of random graphs, we can gain insights into the structure and dynamics of complex networks, and how they differ from more structured networks with community structure or other correlations. Our study cases will be:

- Random symmetric matrix, with entries drawn from a Gaussian distribution, which can be used to model the adjacency matrix of an E-R random graph.
- E-R matrix, which is a specific case of a random symmetric matrix where the entries are binary (0 or 1) and represent the presence or absence of links between nodes in an E-R random graph.
- Hi-C matrix, which is a specific case of a random symmetric matrix that represents the contact frequencies between different regions of the genome in a Hi-C experiment. The spectral properties of Hi-C matrices can provide insights into the 3D organization of the genome and how it relates to gene regulation and other cellular processes.

eigenvalue spectrum of network

For a random symmetric $N \times N$ matrix with elements sampled from a Gaussian pdf $N(0, \sigma)$ the eigenvalue spectrum has the following form. Real spectrum: semicircle centered around origin with a radius depending on node number N and Gaussian SD σ :

$$\rho(\lambda) = \frac{1}{2\pi N \sigma^2} \sqrt{4N\sigma^2 - \lambda^2}, \quad |\lambda| < 2\sigma\sqrt{N} \dots$$

ALL normalized eigenvector components are distributed as a Gaussian pdf: ALL eigenvectors are EQUAL and “poorly informative”

There will be no subsets of nodes particularly emerging from components of our eigenvectors, as they are all equally distributed. This is a key feature of random graphs, where there is no specific structure or correlation in the network, and the eigenvectors do not provide any meaningful information about the structure of the graph.

From the central limit theorem, if we have set of randomly distributed variables, their sum will tend to a Gaussian distribution, regardless of the original distribution of the variables. In the case of random graphs, the eigenvectors are linear combinations of the entries of the adjacency matrix, which are randomly distributed. Therefore, the components of the eigenvectors will also be randomly distributed and will tend to a Gaussian distribution due to the central limit theorem. This means that all eigenvectors will be equally distributed and will not provide any meaningful information about the structure of the graph, which is a key feature of random graphs.

E-R Spectrum

If we take an E-R networks with 0-1 links maximally distributed in a random way, wigner semicircle law will still hold, but with a different radius depending on the link density. It is not exactly the gaussian orthogonal ensemble, but it is still a random symmetric matrix with entries that are randomly distributed. The eigenvalue spectrum of an E-R random graph will still follow the Wigner semicircle law, but the radius of the semicircle will depend on the link density of the graph, and we have a big eigenvalue $\lambda_{\max}$ that emerges from the bulk of the spectrum, which is related to the presence of a giant component in the graph. The presence of this large eigenvalue is a key feature of E-R random graphs and distinguishes them from more structured networks, where there may be multiple large eigenvalues corresponding to different communities or other structures in the graph.

The **maximum eigenvalue** of the adjacency matrix of a network is bound by the average degree $\langle k \rangle$ and the largest degree $k_{\max}$ of the corresponding network as follows:

$$\langle k \rangle \leq \lambda_{\max} \leq k_{\max}.$$

The largest eigenvalue has emerged as a key quantity for the study of a variety of systems and processes, such as virus spread, synchronization of coupled oscillators and stability of couplings in the brain.

Appendices

A | Notation and Conventions

B | Computations and further developments